F1 RACING ANALYTICS

Tableau Project – Complete Sheet & Dashboard Guide

A practical guide to the worksheets discussed in this project, including recommended fields, chart types, calculations, and dashboard layout.

Project	Formula 1 Racing Analysis
Tool	Tableau
Worksheets discussed	18 (some are optional or overlapping)

Important: Field names can differ slightly in your dataset. Use the matching field from your own Data pane. Before final submission, verify that each worksheet uses the correct table and that relationships join matching keys (for example, Season to Season—not Season to Driver).

1. Worksheet plan

No.	Worksheet	Fields / shelves	Chart	Notes
1	Team Championship Points	Team + SUM(Points)	Bar chart	Use the team and points fields from championship_standings.
2	Team Championship Point (alternate view)	Team + SUM(Points)	Bar chart / highlight table	May overlap with Sheet 1; keep only if it shows a different comparison.
3	Driver Championship	Driver (Champion) + SUM(Champion)	Bar chart	Driver-wise championship title count; sort descending.
4	Season-wise Champion	Season + Driver (Champion)	Text table / highlight table	Show champion name by season. A measure is not required for a simple name-by-season view.
5	Team Performance by Points	Team + SUM(Points)	Bar chart	Compare total points by team; sort descending.
6	Race Results Count	Driver + COUNT(Race ID)	Bar chart	Counts recorded race-result rows; not necessarily race wins.
7	Circuit-wise Race Analysis	Circuit + COUNT(Race ID)	Bar chart	Count races/results by circuit; confirm whether Race ID is unique per race.
8	Driver Podium Analysis	Driver + SUM(Podiums)	Bar chart	Use podium total field from championship_standings, if it represents total podiums.
9	Fastest Lap Time by Driver	Driver + MIN(Lap Time S)	Bar chart	From lap_intelligence. Lower time is faster; sort ascending.
10	Tyre Compound Usage	Tyre Compound + COUNT(Lap Number)	Bar chart	From lap_intelligence; counts lap records by compound.
11	Average Lap Time by Tyre	Tyre Compound + AVG(Lap Time S)	Bar chart	Lower average time is faster; compare like-for-like conditions if possible.
12	Pit Stop Analysis	Driver + COUNT(Pit Stop)	Bar chart	Only if a pit_stops table/field exists. Otherwise skip this sheet.
13	Sector Performance	Driver + AVG(Sector 1 S)	Bar chart	Can duplicate for Sector 2 S and Sector 3 S, or make a combined comparison.
14	Average Finishing Position	Driver + AVG(Position)	Bar chart	From race_results. Lower average position is better; exclude non-finish/invalid values only if appropriate.
15	Race Status Distribution	Status + COUNT(Race ID)	Bar chart	From race_results; shows finished, retired, etc. Group statuses if there are many.
16	Pole Positions by Driver	Driver + SUM(Pole Positions)	Bar chart	From championship_standings, if the field exists.
17	Driver Race Wins	Driver + COUNT(Race ID)	Bar chart	Only call this race wins if you filter Position = 1 (or use a verified winner flag).
18	Season-wise Race Performance	Season + COUNT(Race ID), Color: Driver	Stacked bar chart	No filter: this is all recorded race results, not winners only.

Note: Sheets 1 and 5 may be similar, and Sheets 6 and 17 overlap unless Sheet 17 is filtered to Position = 1. It is better to submit fewer distinct, accurate views than many repeated charts.

2. How to build each worksheet in Tableau

Create a worksheet

Click the New Worksheet icon at the bottom and rename the tab with a clear title.

Choose the correct table

In the Data pane, locate the field under the relevant table (for example, lap_intelligence or race_results).

Place fields

Drag the dimension to Columns and the measure to Rows. Drag the same measure to Label if you want values displayed on bars.

Choose a chart

Use the Marks card or Show Me. Bar charts are suitable for comparing drivers, teams, and tyre compounds.

Sort

Use descending order for totals/counts and ascending order for time or finishing-position measures where smaller is better.

Format

Add a clear title, readable labels, sensible number formatting, and avoid too many different colors.

3. Important Tableau calculations and meanings

Expression	Meaning
SUM(Points)	Adds points across records. Use for total team/driver points when the data is at the right level.
COUNT(Race ID)	Counts non-null Race ID records. It may count result rows, not unique races; use COUNTD(Race ID) if you need distinct races.
MIN(Lap Time S)	Smallest recorded lap time for each driver; lower is faster.
AVG(Lap Time S)	Average recorded lap time; lower is faster, but compare under similar conditions.
AVG(Speed Trap Kmh)	Average recorded speed-trap speed, if this field is available.
Position = 1	Restricts race results to first-place finishers. Without this filter, the chart is not a race-wins chart.

Filtering Position = 1: Drag Position from the race_results table to the Filters shelf, select the value 1 (or the exact first-place value used in your data), then click OK. If you choose not to use a filter, rename the chart to Race Results Count/Performance rather than Race Wins.

4. Data relationships – avoid mismatch errors

When Tableau shows a data type mismatch, the relationship is likely connecting fields with different types or unrelated meanings. Match both the meaning and the data type.

Good relationship example	Avoid
race_results.Season ↔ championship_standings.Season	race_results.Season ↔ championship_standings.Driver (Champion)
Driver ↔ Driver (if values use the same naming convention)	Season ↔ Driver
Number ↔ Number; String ↔ String; Date ↔ Date	Number ↔ String without converting/cleaning appropriately

Before creating relationships, confirm that the columns contain the same kind of information. If one field is numeric and the other is text, use the correct matching field or carefully convert the data type only when the values genuinely represent the same thing.

5. Suggested final dashboard

Create one dashboard after the worksheets are checked. A clean layout could be:

Top row	Dashboard title: F1 Racing Analytics | KPI cards: Total Races, Drivers, Teams, Circuits
Middle left	Driver Championship + Driver Race Wins
Middle right	Team Performance + Season-wise Champion
Bottom left	Fastest Lap Time + Average Lap Time by Tyre
Bottom right	Podium Analysis + Race Status Distribution
Filters	Season, Driver, Team (only if the relevant fields/relationships work correctly)

Dashboard tips

• Use a consistent color palette and readable font sizes.

• Keep chart titles short and descriptive.

• Use filters only when they affect the intended charts.

• Avoid duplicate charts that communicate the same information.

• Check totals against the source data before presenting.

6. Final project checklist

■ All worksheet names are clear and spelling is correct.

■ Each sheet uses the correct source table and field.

■ No relationship connects unrelated fields.

■ Counts are understood (COUNT vs COUNTD).

■ Race wins use Position = 1; otherwise label as race results/performance.

■ Fastest lap uses MIN(Lap Time S), not SUM.

■ Average lap/sector times use AVG, not SUM.

■ Pit stop sheet is included only if pit-stop data exists.

■ Dashboard filters and titles work correctly.

■ Save the Tableau workbook (.twb or .twbx) along with this PDF.

Prepared as a project guide from the worksheet plan discussed. Verify field names and results against your actual dataset before submission.